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Exact Coherent Structures in Fluid Turbulence

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Abstract

We consider the incompressible Navier–Stokes equations body-forced by $\sin 4y\hat{\mathbf{x}}$ on the torus $(x, y) \in [0, 2\pi]^2$, with the aim of extracting unstable recurrent flows from high-dissipation regions. Each recurrent flow describes a closed cycle within the state space and, when considered collectively, may provide a framework for describing long-time turbulent dynamics and statistics. In particular, unstable periodic orbits (UPOs) and relative periodic orbits (RPOs) represent recurrent behaviour within the flow, capturing recurring patterns of evolution within the turbulent dynamics. Near-recurrent episodes are identified from DNS using the modified recurrence function R_K , and subsequently converged to exact invariant solutions using the Newton–GMRES–Hookstep method. High-dissipation regions, corresponding to extreme bursting events, contain fewer close recurrences, making them more difficult to identify using the recurrence approach. A DNS trajectory of duration 10^4 was analysed, producing 445 initial guesses, of which 14 converged to eight unique invariant solutions.

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Chapter 1

Introduction

Turbulent flows are characterised by complex and chaotic dynamics. From a dynamical-systems perspective, their evolution can be represented as trajectories through a high-dimensional state space. Embedded within this state space are special solutions of the governing equations, known as invariant solutions or exact coherent structures. Turbulent trajectories can pass close to these solutions before moving away from their neighbourhood, suggesting that unstable invariant solutions may play an important role in organising turbulent dynamics. In particular, unstable periodic orbits (UPOs) and relative periodic orbits (RPOs) represent recurrent behaviour within the flow and, when considered collectively, may provide a framework for describing long-time turbulent dynamics and statistics (Redfern, Lazer, and Lucas 2024). The use of periodic orbits to study chaotic dynamics dates back to Poincaré’s pioneering work on the three-body problem in the 1880s (Callandreau 1892; Ruelle 1978; MacKay and Meiss 1987; Martin C Gutzwiller and M. Gutzwiller 1990). This perspective was subsequently extended to turbulent flows, and Chandler and Kerswell (2013) developed a systematic approach for extracting recurrent solutions from turbulent two-dimensional Kolmogorov flow by identifying near recurrences in DNS trajectories and refining them using the Newton–GMRES–Hookstep (NGh) method.

The two-dimensional Kolmogorov flow consists of the incompressible Navier–Stokes equations on a periodic domain, driven by a steady, monochromatic body force acting at a large spatial scale. This flow was chosen because its simple forcing and periodic domain make it computationally inexpensive and well suited for spectral methods. It also has physical relevance, having been experimentally realised using magnetohydrodynamic forcing (Bondarenko, Gak, and Dolzhanskii 1979) and in soap-film experiments (Burgess et al. 1999). Kolmogorov flow has been studied in a wide range of mathematical and experimental contexts, a more detailed overview is given by Chandler and Kerswell (2013). In this work, the forcing wavenumber is set to $n = 4$ and the Reynolds number to $\text{Re} = 40$. For $n = 4$, the laminar state becomes unstable above $\text{Re} \approx 10$, while at $\text{Re} = 40$ the flow

exhibits sustained spatiotemporal chaos. This parameter regime has been widely studied in the context of invariant solutions, providing a basis for comparison with previous studies.

The turbulent trajectory occasionally undergoes short-lived episodes of unusually high energy dissipation. A key limitation identified by Redfern, Lazer, and Lucas (2024) is that the recurrent-flow analysis of Chandler and Kerswell (2013) did not recover recurrent flows associated with the most extreme high-dissipation bursts. This raises the question of how effectively recurrent-flow analysis can identify invariant solutions within these less frequently visited regions of the turbulent dynamics. Identifying recurrent structures associated with these extreme events therefore remains an active area of research. Recent studies have been motivated by the identification of UPOs associated with high-dissipation dynamics (Page et al. 2024) and have demonstrated that alternative approaches can identify a more diverse set of recurrent solutions, including UPOs and RPOs associated with higher-dissipation bursting events (Redfern, Lazer, and Lucas 2024). We too focus our efforts on obtaining exact recurrent structures embedded within the turbulent flow, with particular attention given to structures associated with regions of high energy dissipation.

Research is organised as follows: Chapter 2 introduces the two-dimensional Kolmogorov flow and formulates the governing equations, including their non-dimensionalisation, vorticity formulation, symmetries, and the flow measures used throughout the analysis. Chapter 3 describes the numerical methodology, beginning with the pseudospectral time-stepping scheme, followed by the detection of near recurrences and the extraction of exact recurrent solutions using the Newton–GMRES–Hookstep method. Chapter 4 presents and discusses the recurrent structures obtained at $Re = 40$. It first examines the turbulent trajectory and the identification of near recurrences, with particular attention to high-dissipation events, before presenting and classifying the converged invariant solutions and comparing them with previously reported solutions. Finally, Chapter 5 summarises the main findings and discusses directions for future work.

Chapter 2

Problem Formulation

This chapter introduces the governing equations for two-dimensional Kolmogorov flow. We begin with the dimensional incompressible Navier–Stokes equations subject to Kolmogorov forcing before deriving their nondimensional form. The equations are then reformulated in terms of the vorticity, resulting in a scalar evolution equation (2.9) that forms the basis of the pseudospectral solver developed in Chapter 3.

For a Newtonian fluid of constant density ρ , the dimensional incompressible Navier–Stokes equations with Kolmogorov forcing are

$$\frac{\partial \mathbf{u}^*}{\partial t^*} + (\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* = -\frac{1}{\rho} \nabla^* p^* + \nu \nabla^{*2} \mathbf{u}^* + \chi \sin(2\pi n y^*/L_y) \hat{\mathbf{x}}, \quad (2.1)$$

$$\nabla^* \cdot \mathbf{u}^* = 0. \quad (2.2)$$

where $\mathbf{u}^* = u^* \hat{\mathbf{x}} + v^* \hat{\mathbf{y}}$ is the dimensional velocity field, ρ is the density, ν is the kinematic viscosity, n is an integer describing the scale of the monochromatic Kolmogorov forcing, and χ is the forcing amplitude per unit mass of fluid over the doubly periodic domain $[0, L_x] \times [0, L_y]$, with $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ denoting the unit vectors in the x^* and y^* directions, respectively (In this chapter only, the superscript $*$ denotes a dimensional quantity).

2.1 Non-dimensionalisation

Following Chandler and Kerswell (2013) we introduce the dimensionless variables

$$x = \frac{x^*}{L}, \quad y = \frac{y^*}{L}, \quad \mathbf{u} = \frac{\mathbf{u}^*}{U}, \quad t = \frac{t^*}{T}, \quad p = \frac{p^*}{\rho U^2}, \quad (2.3)$$

where $L = L_y/2\pi$, $U = \sqrt{\chi L}$, $T = L/U$. Throughout this project we take $L_x = L_y = 2/\pi$, so the dimensionless computational domain is $[0, 2\pi] \times [0, 2\pi]$. The incompressibility

condition becomes

$$\nabla \cdot \mathbf{u} = 0. \quad (2.4)$$

Substituting into (2.1) and dividing by U^2/L gives

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{\nu}{UL} \nabla^2 \mathbf{u} + \frac{\chi L}{U^2} \sin(ny) \hat{\mathbf{x}}. \quad (2.5)$$

Defining the Reynolds number where the Reynolds number is defined by

$$Re := \frac{\sqrt{\chi}}{\nu} \left(\frac{L_y}{2\pi} \right)^{3/2}, \quad (2.6)$$

and noting that $U^2 = \chi L$, the governing equations become

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u} + \sin(ny) \hat{\mathbf{x}}, \quad (2.7)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2.8)$$

2.2 Vorticity Equation

For two-dimensional Kolmogorov flow, it is convenient to work in the vorticity–streamfunction formulation. Taking the curl of (2.7) eliminates the pressure and reduces the Navier–Stokes equations to a single scalar vorticity equation. Since $\boldsymbol{\omega} = \nabla \times \mathbf{u} = \omega \hat{\mathbf{z}}$ and $\nabla \times [\sin(ny)\hat{\mathbf{x}}] = -n \cos(ny)\hat{\mathbf{z}}$, the governing equation becomes

$$\frac{\partial \omega}{\partial t} + J(\psi, \omega) = \frac{1}{Re} \nabla^2 \omega - n \cos(ny), \quad (2.9)$$

where the Jacobian is defined by $J(\psi, \omega) = \partial\psi/\partial y \partial\omega/\partial x - \partial\psi/\partial x \partial\omega/\partial y$. For incompressible flow, the velocity is expressed in terms of the streamfunction as $\mathbf{u} = (\partial\psi/\partial y, -\partial\psi/\partial x)$ so that the velocity is recovered from the vorticity by solving the Poisson equation for ψ .

2.3 Symmetries

The two-dimensional Kolmogorov flow possesses both discrete and continuous symmetries. The first is the *shift-and-reflect* symmetry,

$$\mathcal{S} : [u, v, \omega](x, y) \mapsto [-u, v, -\omega] \left(-x, y + \frac{\pi}{n} \right), \quad (2.10)$$

which combines a reflection in the x -direction with a shift of half a forcing wavelength in the y -direction ($\mathbf{u} := u \hat{\mathbf{x}} + v \hat{\mathbf{y}}$, $\omega := \partial v/\partial x - \partial u/\partial y$). Since the forcing has n wave-

lengths across the domain, repeated applications of S generate a finite cyclic symmetry group of order $2n - 1$. The flow also has a rotation-by- π symmetry,

$$\mathcal{R} : [u, v, \omega](x, y) \mapsto [-u, -v, \omega](-x, -y). \quad (2.11)$$

Finally, the governing equations are invariant under continuous translations in the stream-wise direction,

$$\mathcal{T}_l : [u, v, \omega](x, y) \mapsto [u, v, \omega](x + l, y), \quad \text{for} \quad 0 \leq l < \frac{2\pi}{\alpha}, \quad (2.12)$$

where α is the streamwise wavenumber. These transformations leave the governing equations invariant. That is, applying any of the above symmetry operations to a solution produces another solution of the same equations. Consequently, symmetry-related states are physically equivalent even though they may have different representations in Fourier space.

2.4 Flow measures

In order to analyse and compare the recurrent flows and turbulent statistics, the following diagnostic quantities are introduced. The total kinetic energy, energy dissipation rate and energy input rate are defined by

$$E(t) := \frac{1}{2} \langle \mathbf{u}^2 \rangle_V, \quad D(t) := \frac{1}{\text{Re}} \langle |\nabla \mathbf{u}|^2 \rangle_V, \quad I(t) := \langle u \sin(ny) \rangle_V, \quad (2.13)$$

where $\langle \cdot \rangle_V$ denotes the volume average defined by

$$\langle \cdot \rangle_V := \frac{1}{4\pi^2} \int_0^{2\pi} \int_0^{2\pi} (\cdot) \, dx \, dy. \quad (2.14)$$

The energy budget satisfies $\frac{dE}{dt} = I - D$, meaning that any steady state, and the turbulent time average, must satisfy $D = I$. For Kolmogorov flow, the laminar solution is maintained by an exact balance between forcing and dissipation. It is given by

$$\mathbf{u}_{\text{lam}} = \frac{\text{Re}}{n^2} \sin(ny) \hat{\mathbf{x}}, \quad \omega_{\text{lam}} = \frac{\text{Re}}{n} \cos(ny), \quad (2.15)$$

with corresponding energy and dissipation rate

$$E_{\text{lam}} = \frac{\text{Re}^2}{4n^4}, \quad D_{\text{lam}} = \frac{\text{Re}^2}{2n^2}. \quad (2.16)$$

For the case $n = 4$, the laminar solution loses stability at approximately $\text{Re} \approx 10$, and by $\text{Re} = 40$ the flow exhibits sustained spatiotemporal chaos (Chandler and Kerswell 2013).

Chapter 3

Methodology

3.1 Time stepping code

We implement two-dimensional fully de-aliased pseudo-spectral code developed by Chandler and Kerswell (2013). The Crank–Nicolson method was used for the viscous terms, and Heun’s method (Euler predictor method) was used for the nonlinear and forcing terms so that only one state vector was required to accurately proceed to the next timestep. The vorticity field is represented by the Fourier series

$$\omega(x, y, t) = \sum_{k_x} \sum_{k_y} \Omega_{k_x, k_y}(t) e^{i(k_x x + k_y y)}. \quad (3.1)$$

Here the wavenumbers are

$$k_x = \frac{2\pi m}{L_x}, \quad k_y = \frac{2\pi l}{L_y}, \quad m, l \in \mathbb{Z}, \quad (3.2)$$

as required by the periodicity of the domain. A key advantage of the Fourier representation is that spatial differentiation reduces to multiplication by the corresponding wavenumber in spectral space, for example

$$\frac{\partial \omega}{\partial x} = \sum_{k_x} \sum_{k_y} i k_x \Omega_{k_x, k_y}(t) e^{i(k_x x + k_y y)}. \quad (3.3)$$

If $\Omega(\mathbf{k}) := \mathcal{F}(\omega(\mathbf{x}))$ denotes the Fourier transform of ω , with $\mathbf{k} = (k_x, k_y)$, then the viscous diffusion term in spectral space is

$$\mathcal{F}\left(\frac{1}{\text{Re}} \nabla^2 \omega\right) = -G\Omega, \quad (3.4)$$

where

$$G(k_x, k_y) := \frac{k_x^2 + k_y^2}{\text{Re}} = \frac{|\mathbf{k}|^2}{\text{Re}}. \quad (3.5)$$

If every term in the vorticity equation were linear, the entire computation could be performed in spectral space. However, the nonlinear advection term contains products of fields, which correspond to convolutions in Fourier space and are computationally expensive to evaluate directly. Instead, the nonlinear term is evaluated in physical space, where the required products can be computed pointwise. The nonlinear term can be written in conservative form as

$$\mathbf{u} \cdot \nabla \omega = \nabla \cdot (\omega \mathbf{u}), \quad (3.6)$$

since $\nabla \cdot \mathbf{u} = 0$. In components, $\mathbf{u} \cdot \nabla \omega = \partial(u\omega)/\partial x + \partial(v\omega)/\partial y$. The velocity and its derivatives are obtained in spectral space, after which the products $u\omega$ and $v\omega$ are evaluated in physical space. The resulting nonlinear term is then transformed back to spectral space. We compute nonlinear advection and forcing in spectral space using

$$f(\Omega) := -i(k_x \mathcal{F}(u\omega) + k_y \mathcal{F}(v\omega)) - \frac{n}{2} \delta_{k_x,0} (\delta_{k_y,n} + \delta_{k_y,-n})$$

where the Kronecker delta functions indicate that the forcing is non-zero only at these Fourier modes. The vorticity equation (2.9) in spectral space becomes

$$\frac{\partial \Omega}{\partial t} = -G\Omega + f(\Omega), \quad (3.7)$$

Time integration is performed using a two-stage semi-implicit scheme. The viscous diffusion term is treated implicitly with the Crank–Nicolson method, while the nonlinear term is evaluated explicitly. An intermediate solution Ω^* is first computed using a forward Euler step, giving

$$\frac{\Omega^* - \Omega^i}{\Delta t} = -\frac{G}{2} (\Omega^* + \Omega^i) + f(\Omega^i). \quad (3.8)$$

The solution is then advanced using Heun’s method for the nonlinear term, which gives

$$\frac{\Omega^{i+1} - \Omega^i}{\Delta t} = -\frac{G}{2} (\Omega^{i+1} + \Omega^i) + \frac{1}{2} [f(\Omega^*) + f(\Omega^i)]. \quad (3.9)$$

Here, the superscript denotes the discrete time-step index. With de-aliasing, a resolution of $N_x \times N_y$ corresponds in practice to the vorticity representation

$$\omega(x, y, t) = \sum_{k_x=-N_x/3}^{N_x/3-1} \sum_{k_y=-N_y/3}^{N_y/3-1} \Omega_{jl}(t) e^{i(k_x x + k_y y)}, \quad (3.10)$$

with $\Omega_{00} = 0$. A spectral mask is employed so that $\Omega_{jl} = 0$ for wavenumbers outside a specified domain $\Sigma = \{(k_x, k_y) : k_x^2 + k_y^2 \leq (N_x/3)^2\}$. For $N_x = N_y = 128$, this corresponds to a circular region in Fourier space with an area of $\pi (N_x/3)^2 \approx 5720$, containing 5524 retained modes. Since the vorticity field is real, its Fourier coefficients satisfy the

Hermitian symmetry $\Omega(-k_x, -k_y) = \overline{\Omega(k_x, k_y)}$. Thus, the coefficients at opposite wavevectors are not independent, and only one half of the retained Fourier modes needs to be stored explicitly.

3.2 Detection of near recurrences

The identification of exact coherent structures from turbulent DNS data first requires detecting near recurrences, where the flow nearly revisits a previous state. These recurrent episodes provide initial guesses for Newton-based methods, which refine them into periodic or relative periodic orbits as will be explained.

A recurrence is an episode in which the flow at time $t + T$ returns approximately to its state at time t , up to the continuous streamwise translation $\mathcal{T}_s$ and the discrete shift symmetry $\mathcal{S}^{2m}$. For an exact recurrent flow, this condition is written as

$$\mathcal{T}_s \mathcal{S}^{2m} \omega(x, y, t + T) := \omega\left(x + s, y + \frac{2\pi m}{n}, t + T\right) = \omega(x, y, t), \quad (3.11)$$

for some shift $0 \leq s < 2\pi$, discrete symmetry index $m \in \{0, 1, \dots, n-1\}$, and period $T > 0$. Different choices of s , m , and T define different invariant solutions. Periodic orbits satisfy $s = m = 0$, while travelling waves satisfy $m = 0$ and $s = cT$. Equilibria correspond to $s = m = 0$ for arbitrary T , whereas relative periodic orbits have $s \neq 0$ and/or $m \neq 0$. These classifications require care for symmetry-invariant states; for example, the laminar equilibrium is invariant under these symmetries, so non-zero s or m does not produce a distinct solution.

To identify candidate invariant solutions, Chandler and Kerswell (2013) introduced the recurrence function

$$R(t, T) := \min_{\substack{0 \leq s < 2\pi \\ m \in \{0, \dots, n-1\}}} \frac{\sum_{\mathbf{k}} |\Omega_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \boldsymbol{\delta}_m(s)} - \Omega_{\mathbf{k}}(t - T)|^2}{\sum_{\mathbf{k}} |\Omega_{\mathbf{k}}(t)|^2}, \quad (3.12)$$

with $\boldsymbol{\delta}_m(s) = (s, \frac{2\pi m}{n})$. Small values of $R(t, T)$ indicate near recurrences. Minimizing over the symmetry parameters, particularly continuous streamwise shift s , for every candidate recurrence is computationally expensive. Instead, a symmetry-reduced recurrence function based on the method of slices was used (Budanur et al. 2015). Let $\phi_{(1,0)}(t)$ denote the phase of the $(1, 0)$ Fourier mode. We use the first Fourier mode slice $\bar{\phi}(t) = \phi_{(1,0)}(t)$. We further modify the recurrence function by weighting each Fourier mode according to $|\mathbf{k}|^2 = k_x^2 + k_y^2$. Since viscous dissipation is weighted towards small-scale, high-wavenumber

structures, this places greater emphasis on their recurrence. We apply this measure in regions of high dissipation, where near-recurrent states can then be distinguished more clearly, further details are presented in the following chapter. The modified recurrence function is

$$R_K(t, T) := \frac{\sum_{\mathbf{k}} |\mathbf{k}|^2 \left| \Omega_{\mathbf{k}}(t) e^{-i\bar{\phi}(t)} - \Omega_{\mathbf{k}}(t - T) e^{-i\bar{\phi}(t-T)} \right|^2}{\sum_{\mathbf{k}} |\mathbf{k}|^2 |\Omega_{\mathbf{k}}(t)|^2}. \quad (3.13)$$

The phase factors place each state on the first-Fourier-mode slice, $\bar{\phi}(t) = 0$, thereby removing the continuous streamwise translation.

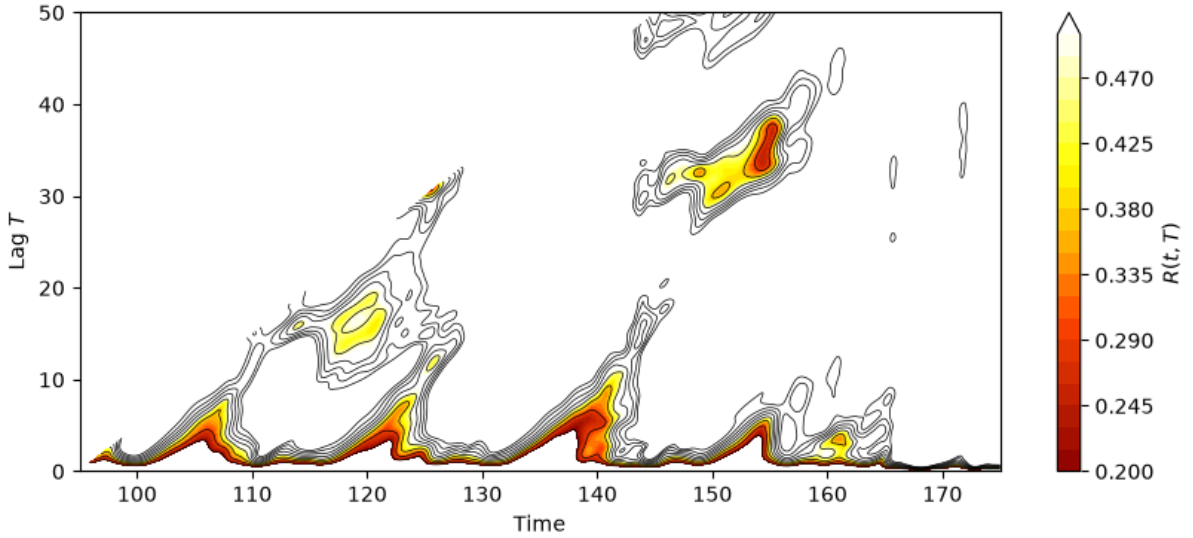


Figure 3.1: Example of $R_K(t, T)$ at $Re = 40$, contoured over $t \in [95, 175]$ (x-axis) and $T \in [0, 50]$ (y-axis).

Figure 3.1 shows a typical recurrence landscape, illustrating the variation of $R_K(t, T)$ with the initial time t and recurrence period T . Darker regions correspond to smaller values of R_K , indicating near recurrences. Since any flow state remains close to itself over sufficiently short time intervals, trivial near recurrences are always present for small T . When computing Figure 3.1, we retained the DNS history over $t \in [95, 175]$. The missing data in the region $t \in [95, 145]$ above the line with slope 1 arise because insufficient history was stored to evaluate $R_K(t, T)$ in this region. We will return to this observation in Section 4.1. Previous studies have shown a clear correlation between the period of recurrent flows and their mean dissipation: high-dissipation UPOs are almost exclusively short-period ($T < 10$) (Redfern, Lazer, and Lucas 2024), whereas longer-period orbits are associated with more regular, low-dissipation dynamics. Guided by these observations, and noting that the longest high-dissipation UPOs reported have periods below $T = 35$, the recurrence search was performed over the interval $2 < T \leq 35$.

3.3 UPO extraction method: Newton-GMRES-hookstep

Once a near recurrence has been identified using the criterion described above, we sought to determine whether an exact recurrent flow existed in its neighbourhood in phase space. This is achieved by applying a high-dimensional root-finding algorithm for a state vector that completely specifies the recurrent solution, given by

$$\mathbf{X} = \begin{bmatrix} \boldsymbol{\Omega} \\ s \\ T \end{bmatrix}. \quad (3.14)$$

where $\boldsymbol{\Omega}$ is the vector of Fourier coefficients Ω_{jl} describing the vorticity field, s is the spatial shift associated with a relative periodic orbit, and T is the recurrence period. The discrete symmetry index m is held fixed throughout the Newton iteration, as it cannot be adjusted continuously. To formulate the Newton–Raphson algorithm, it is convenient to introduce the infinitesimal generators of translations in the x - and y -directions, denoted by T_x and T_y , respectively,

$$T_x \omega(x, y, t) \rightarrow \frac{\partial \omega}{\partial x} = \sum_{l=-N_y/3}^{N_y/3-1} \sum_{j=-N_x/3}^{N_x/3-1} i \alpha j \Omega_{jl}(t) e^{i(\alpha j x + l y)}, \quad (3.15)$$

$$T_y \omega(x, y, t) \rightarrow \frac{\partial \omega}{\partial y} = \sum_{l=-N_y/3}^{N_y/3-1} \sum_{j=-N_x/3}^{N_x/3-1} i l \Omega_{jl}(t) e^{i(\alpha j x + l y)}. \quad (3.16)$$

as they act in spectral space $T_x \boldsymbol{\Omega} \rightarrow \boldsymbol{\Omega}_x$ and $T_y \boldsymbol{\Omega} \rightarrow \boldsymbol{\Omega}_y$. Then, the recurrence condition (3.11) in spectral space becomes

$$\mathbf{F}(\boldsymbol{\Omega}_0, s, T; m) = \exp\left(s T_x + \frac{\pi}{2} m T_y\right) \tilde{\boldsymbol{\Omega}}(\boldsymbol{\Omega}_0, T) - \boldsymbol{\Omega}_0 = \mathbf{0}, \quad (3.17)$$

where $\boldsymbol{\Omega}_0 = \boldsymbol{\Omega}(t)$ and $\tilde{\boldsymbol{\Omega}}(\boldsymbol{\Omega}_0, T) = \boldsymbol{\Omega}(t + T)$. Given an initial guess $\mathbf{X}_0 = (\boldsymbol{\Omega}_0, s_0, T_0)^T$, the Newton iteration seeks a correction such that the updated approximation $\mathbf{X}_0 + \delta \mathbf{X} = (\boldsymbol{\Omega}_0 + \delta \boldsymbol{\Omega}, s_0 + \delta s, T_0 + \delta T)^T$ satisfies the linearized recurrence condition

$$\frac{\partial \mathbf{F}}{\partial \boldsymbol{\Omega}_0} \delta \boldsymbol{\Omega} + \frac{\partial \mathbf{F}}{\partial s} \delta s + \frac{\partial \mathbf{F}}{\partial T} \delta T = -\mathbf{F}(\boldsymbol{\Omega}_0, s_0, T_0; m). \quad (3.18)$$

This forms a system of $\dim(\boldsymbol{\Omega})$ equations for $\dim(\boldsymbol{\Omega}) + 2$ unknowns. Since a periodic or relative periodic orbit may be represented starting from any state along the orbit, and a relative periodic orbit may also be shifted arbitrarily in the streamwise direction, two

phase conditions are imposed to remove these ambiguities:

$$\left\langle \delta \mathbf{\Omega}, \frac{\partial \mathbf{\Omega}}{\partial t} \right\rangle = 0, \quad \text{and} \quad \langle \delta \mathbf{\Omega}, T_x \mathbf{\Omega} \rangle = 0. \quad (3.19)$$

The Newton–Raphson problem is then to solve

$$\begin{bmatrix} \ddots & & & \vdots & \vdots \\ & \frac{\partial \hat{\mathbf{\Omega}}^s}{\partial \mathbf{\Omega}_0} - \mathbf{I} & & T_x \hat{\mathbf{\Omega}}^s & \frac{\partial \hat{\mathbf{\Omega}}^s}{\partial T} \\ & & \ddots & \vdots & \vdots \\ \cdots & (T_x \mathbf{\Omega}_0)^T & \cdots & 0 & 0 \\ \cdots & \left(\frac{\partial \mathbf{\Omega}_0}{\partial t} \right)^T & \cdots & 0 & 0 \end{bmatrix} \begin{bmatrix} \vdots \\ \delta \mathbf{\Omega} \\ \vdots \\ \delta s \\ \delta T \end{bmatrix} = - \begin{bmatrix} \vdots \\ F(\mathbf{\Omega}_0, s_0, T_0; m) \\ \vdots \\ 0 \\ 0 \end{bmatrix}. \quad (3.20)$$

where $\tilde{\mathbf{\Omega}}^s := \exp(sT_x + (1/2)\pi m T_y) \tilde{\mathbf{\Omega}}$ is the ‘back-shifted’ final state, and I denotes the $\dim(\mathbf{\Omega}) \times \dim(\mathbf{\Omega})$ identity matrix. The Newton system can therefore be written in the standard form $\mathbf{A} \delta \mathbf{X} = \mathbf{b}$. The only non-trivial block of the Jacobian is $\partial \mathbf{\Omega}_s / \partial \mathbf{\Omega}_0$. The remaining derivatives $\partial \mathbf{\Omega}_s / \partial T$ and $\partial \mathbf{\Omega}_0 / \partial t$ follow directly from the Navier–Stokes equations.

In practice, the Jacobian matrix $\mathbf{A}$ is too large to be formed or stored explicitly, making a direct solution of $\mathbf{A} \delta \mathbf{X} = \mathbf{b}$ infeasible. Instead, the system is solved iteratively using the Generalized Minimal Residual (GMRES) method, which requires only the action of $\mathbf{A}$ on an arbitrary vector. The action of the Jacobian $\partial \mathbf{\Omega}_s / \partial \mathbf{\Omega}_0$ is evaluated using a forward finite difference

$$\frac{\partial \mathbf{\Omega}_s}{\partial \mathbf{\Omega}_0} \mathbf{y} \approx \frac{\mathbf{\Omega}_s(\mathbf{\Omega}_0 + \varepsilon \mathbf{y}) - \mathbf{\Omega}_s(\mathbf{\Omega}_0)}{\varepsilon}, \quad (3.21)$$

where ε is chosen such that $\|\varepsilon \mathbf{y}\| = 10^{-7} \|\mathbf{\Omega}_0\|$, balancing truncation and round-off errors in double-precision arithmetic. For the Newton–GMRES method, convergence is only guaranteed locally to the solution, so a full Newton step is most reliable when the current iterate is sufficiently close to the solution. Since the initial guess may lie farther from the solution, the local linearisation may not provide a suitable full step. The hookstep therefore constrains the Newton correction to a prescribed trust-region radius, allowing smaller steps when the full Newton step is unsuitable. While Chandler and Kerswell (2013) employ a Hookstep trust-region method based on the GMRES Hessenberg matrix, we instead implement a simple backtracking line search, repeatedly halving the Newton step until the residual decreases, i.e.

$$\delta \mathbf{X} \rightarrow \frac{1}{2} \delta \mathbf{X} \quad \text{until} \quad \|\mathbf{F}(\mathbf{X} + \delta \mathbf{X})\| \leq \|\mathbf{F}(\mathbf{X})\|. \quad (3.22)$$

Chapter 4

Discussion

4.1 Finding recurrent structures

The time-stepping code was initialized with a vorticity field whose values were drawn independently from a zero-mean Gaussian distribution with standard deviation 0.1. The DNS was then integrated forward in time, with the first 500 time units discarded to eliminate transient behaviour. Figure 4.1 shows the normalised energy dissipation rate over the interval $500 \leq t \leq 1500$. For most of the simulation, the dissipation remains in the range $0.06 \lesssim D/D_{\text{lam}} \lesssim 0.14$, interrupted by intermittent bursts of substantially higher dissipation. Throughout this work, a high-dissipation event is defined by $D/D_{\text{lam}} > D_{\text{thresh}}$, where $D_{\text{thresh}} = 0.15$. Since the aim of this work is to identify highly dissipative exact coherent structures, these bursting events are the primary focus of the recurrence analysis.

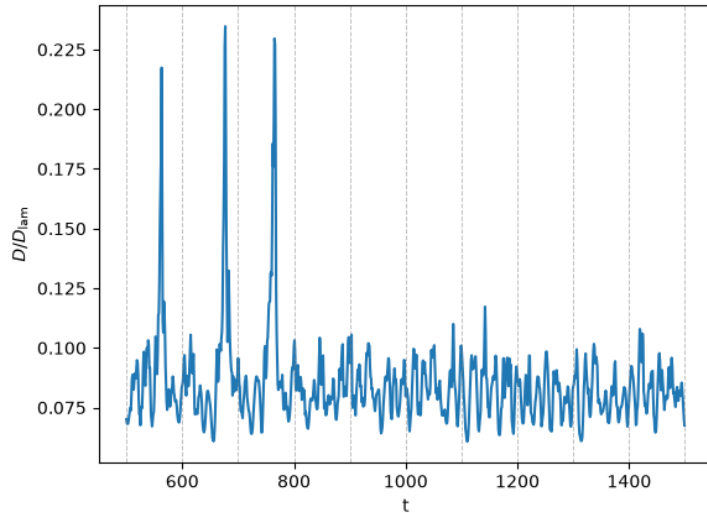


Figure 4.1: Dissipation D/D_{lam} as a function of time over the interval $t \in [500, 1500]$.

Using the recurrence function $R_K(t, T)$ (3.13), Figure 4.2 shows the recurrence landscape over the interval $660 \leq t \leq 800$. This interval was chosen because it contains two high-

dissipation events, $I_1 = (673.5, 679)$ and $I_2 = (761, 767.5)$, together with periods of more typical turbulent behaviour.

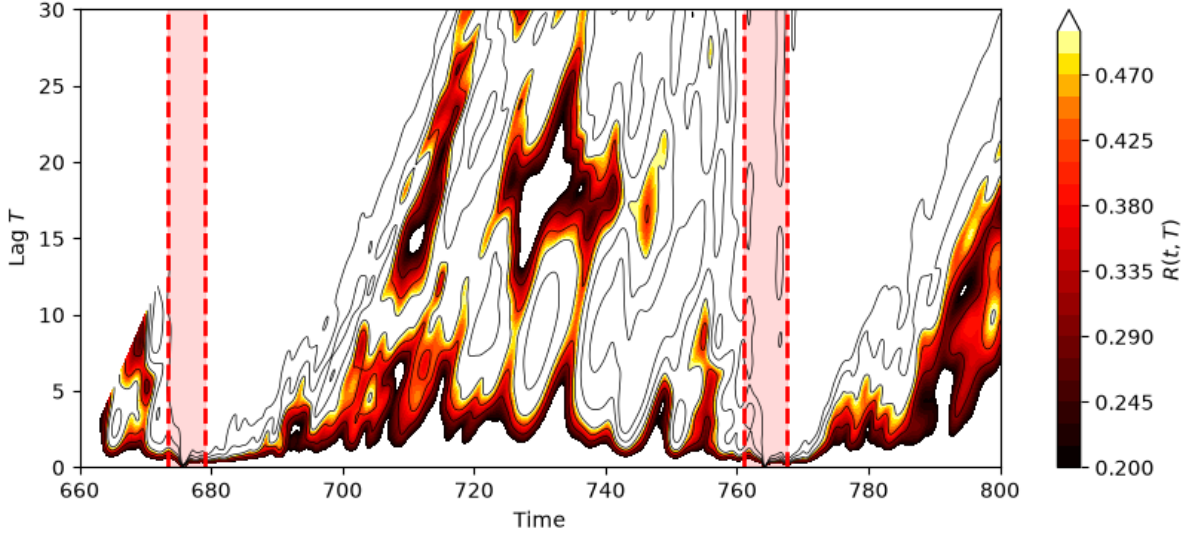


Figure 4.2: Recurrence landscape for $660 \leq t \leq 800$, showing the variation of the recurrence function R_K with time t and recurrence time T .

The recurrence function R_K is noticeably larger during high-dissipation bursts than during the surrounding turbulent dynamics. Consequently, these episodes provide fewer close recurrences from which to construct initial guesses for the Newton–GMRES–Hookstep method. This difficulty is consistent with the challenges encountered in identifying high-dissipation recurrent structures using standard recurrence-based approaches, while recent studies by Page et al. (2024) and Redfern, Lazer, and Lucas (2024) have demonstrated that such structures can be recovered using alternative approaches.

While Chandler and Kerswell (2013) evaluated the recurrence function throughout the DNS trajectory, applying the same approach to the present problem would be computationally inefficient, since the high-dissipation events of interest occupy only a small fraction of the turbulent trajectory. We therefore monitor the dissipation during the DNS and activate the recurrence calculation only when $D > D_{\text{thres}}$. However, beginning to store Ω_{history} only after the threshold is crossed is insufficient, since the high-dissipation intervals are relatively short, as illustrated by I_1 and I_2 , and the stored history would not contain the earlier states required to evaluate $R_K(t, T)$. As discussed above, this would produce a region in the recurrence landscape where R_K cannot be evaluated. To retain the required history efficiently without storing the entire DNS trajectory, we maintain Ω_{history} using a ring buffer.

The recurrence searches were performed over a DNS trajectory of duration 10^4 , using a fixed time step of $\Delta t = 0.01$ (Table 4.1). For the standard thresholds $D_{\text{thres}} = 0.15$ and

$R_{\text{thres}} = 0.70$, each search produced between 42 and 57 initial guesses, with between one and three converging to exact invariant solutions. Increasing the recurrence threshold to $R_{\text{thres}} = 0.85$ substantially increased the number of candidate guesses to 104, but did not improve the number of converged solutions. Conversely, lowering the threshold to $R_{\text{thres}} = 0.60$ produced fewer guesses and one convergences. Increasing the dissipation threshold to $D_{\text{thres}} = 0.20$ was particularly restrictive, yielding only four guesses and no converged solutions. Across all nine searches, 445 initial guesses were generated, of which 14 converged to invariant solutions.

Run	D_{thres}	R_{thres}	Δt	Duration	No. of guesses	No. of convergences
a	0.15	0.70	0.01	10^4	57	1
b	0.15	0.70	0.01	10^4	42	3
c	0.15	0.70	0.01	10^4	53	2
d	0.15	0.70	0.01	10^4	51	1
e	0.15	0.70	0.01	10^4	48	2
f	0.15	0.85	0.01	10^4	104	2
g	0.15	0.6	0.01	10^4	34	1
h	0.20	0.70	0.01	10^4	4	0
i	0.15	0.70	0.01	10^4	52	3

Table 4.1: Summary of recurrence searches. R_{thres} denotes the recurrence threshold, Δt the time step used in the search, and the final two columns give the number of initial recurrence guesses and the number of converged invariant solutions, respectively.

The initial pass through the DNS data to identify near recurrences took approximately 1.5 hours per run on a Google Cloud C4A High-CPU virtual machine (16 vCPUs, 32 GB RAM). The computation slowed considerably during high-dissipation events, $D > D_{\text{thres}}$, when the recurrence search was activated. The convergence of near-recurrent guesses to exact invariant solutions proved to be the primary computational bottleneck, with some individual guesses requiring up to one hour to compute. For comparison, Chandler and Kerswell (2013) considered DNS trajectories extending to 10^5 time units and searched across the full turbulent trajectory, whereas the present work considers 10^4 time units and specifically targets high-dissipation episodes. Consequently, the computational cost in the present work is on the scale of hours for recurrence searches and days for the overall NGh calculations, compared with weeks for generating recurrence guesses and months for their NGh calculations (Chandler and Kerswell 2013).

In Figure 4.3, we plot the typical Newton residuals for several initial guesses, together with those of a number of converged guesses. Here, we refer to *stagnation* as a period during which the Newton residual remains approximately constant over several iterations, with little or no decrease. Contrary to our initial expectation based on the NGh analysis of Farazmand (2016), some of our converged guesses exhibited residual stagnation before eventually converging. In contrast, Farazmand (2016) found that residuals stagnating

above approximately 10^{-5} did not subsequently converge. Furthermore, his converged guesses reached residuals of approximately 10^{-5} within 3–15 iterations, whereas our converged guesses typically required 29–34 iterations to reach a comparable residual level. Although this behaviour was initially unexpected, it may be related to the relatively high R_{thres} used in our recurrence search.

The Newton–GMRES–Hookstep (NGh) implementation was provided as a pre-existing codebase. Several modifications to the termination behaviour were considered to improve the efficiency and robustness of processing multiple initial guesses. Following the behaviour reported by Farazmand (2016), we initially used residual stagnation above 10^{-5} as a criterion for terminating a guess early, allowing the available guesses to be processed more efficiently. However, inspection of the residual histories showed that this criterion could discard potentially successful guesses. In particular, some of the guesses that eventually converged in our analysis exhibited stagnation above 10^{-5} before subsequently reducing their residual and converging. Applying this criterion would therefore have removed these successful cases from consideration.

Furthermore, guesses that had not reached the convergence threshold of 10^{-10} and had not reached the maximum of 50 Newton iterations were terminated when the backtracking line search reduced the Newton step below the minimum step size of 10^{-13} . At this point, the condition in Eq. 3.22 could no longer be satisfied, causing the NGh calculation to terminate for that guess and proceed to the next initial guess. This prevented a failed guess from terminating the program and allowed the remaining guesses to be processed automatically, rather than requiring the failed guess to be skipped manually, as was initially required.

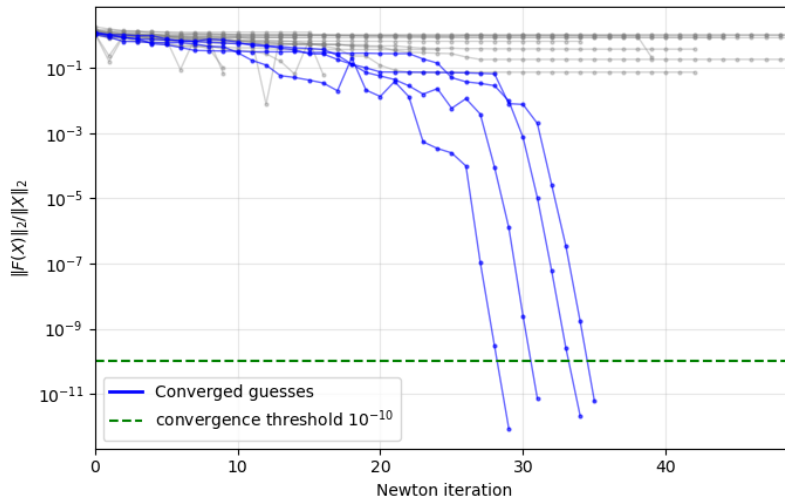


Figure 4.3: Newton residual histories for representative NGh initial guesses. Converged and non-converged cases are shown, with the convergence threshold set to 10^{-10} .

4.2 Recurrent structures found

In this section, we report the recurrent structures identified at $Re = 40$, as summarized in Table 4.2. The first classification tool we employed was to examine the relationship between the energy input I and dissipation D , as shown in Figure 4.4. Equilibrium and travelling-wave solutions appear as single points in the $I - D$ plane, since their energy input and dissipation remain constant. In contrast, periodic and relative periodic orbits trace closed loops in the $I - D$ plane as the solution evolves over one period. To complete the classification, we distinguish between equilibria and travelling waves, and between periodic and relative periodic orbits, by examining their discrete and continuous spatial shifts.

	c	$\overline{D}$	$\overline{E}$	T	S_x	S_y	N	$\sum_{j=1}^N \text{Re}(\lambda_j)$	$\max_j \text{Re}(\lambda_j)$
E_1	-	0.07270	0.56826	-	0	0	3	1.41949	0.51991
E_2	-	0.12717	0.76193	-	0	0	2	0.668209	0.401398
TW_1	0.0197	0.08849	0.69748	-	0.02091	0	4	0.141	0.068
P_1	-	0.11892	0.74766	2.82	0	0	4	0.10528	0.04874
R_1	-	0.15377	0.63258	7.55	0.04626	0	4	0.35666	0.22712
R_2	-	0.1421	0.5983	6.92	-0.04740	0	6	0.96367	0.37197
R_3	-	0.190463	0.62336	4.78	0.24136	0	5	0.83611	0.25333
R_4	-	0.19284	0.60362	4.52	-0.38213	0	7	0.27507	0.95068

Table 4.2: Properties of the converged invariant solutions. Equilibria (E_i), travelling waves (T_i), periodic orbit (P_1), and relative periodic orbits (R_i).

The stability of the invariant solutions is characterised by their Floquet exponents. The number of unstable directions is given by $N = \#\{j : \text{Re}(\lambda_j) > 0\}$. We also report the maximum real part, $\max_j \text{Re}(\lambda_j)$, which measures the growth rate of the most unstable perturbation, and the sum $\sum_{j=1}^N \text{Re}(\lambda_j)$, which provides a measure of the total instability associated with the unstable directions.

In total, we converged 14 initial guesses, yielding eight unique solutions, with TW_1 and P_1 each recovered multiple times. We then compared the converged solutions with invariant solutions previously reported in the literature. Our travelling wave TW_1 shows close agreement with the $\tilde{T}_1$ reported by Chandler and Kerswell (2013), while our periodic orbit P_1 closely matches their $\tilde{P}_2$. Our equilibrium E_2 also shows close agreement with the solution reported by Farazmand (2016). At $Re = 40$, Chandler and Kerswell reported $\tilde{T}_1$ 127 times, while we recovered TW_1 three times, indicating that it was frequently recovered in both searches. In contrast, they reported $\tilde{P}_1$ 143 times and $\tilde{P}_2$ 6 times, while we recovered P_1 six times but did not recover their frequently observed $\tilde{P}_1$.

E_1 , E_2 and TW_1 are illustrated in Figure 4.5; one figure per solution suffices, as some snapshots would appear identical for E_1 and E_2 , while those for TW_1 would differ only by the streamwise shift. More interesting are the periodic and relative periodic orbits, for which the flow state varies depending on where along the orbit the snapshot is taken in the $I - D$ plane. For the relative periodic orbit R_1 , snapshots were taken at times 0, 1.51, 3.02, 4.53, 6.04, 7.40, shown in Figures 4.6, corresponding to images 1–6 in Figure 4.10. The orbit starts and ends at the same point, explaining the resemblance between images 1 and 6. Images 2 and 5 also lie close together in the $I - D$ plane and exhibit similar flow structures. However, since the $I - D$ plot is such a drastic projection of the dynamics, two flows that appear close in this plane are not necessarily close in the full phase space.

Similarly, for the relative periodic orbit R_3 , snapshots were taken at times 0, 1.19, 2.39, 3.58, 4.60, shown in Figure 4.9b, corresponding to images 1–6 in Figure 4.11. In contrast to R_1 , where similarities can be observed between snapshots 1 and 6 and between 2 and 5, the snapshots of R_3 show little visual resemblance to one another, apart from the first and final snapshots, which correspond to nearly the same point on the orbit.

Among the recurrent solutions identified, periodic orbit P_1 exhibits the most visually distinctive dynamics. Snapshots are taken at times 0, 0.56, 1.12, 1.69, 2.25, 2.82, as shown in Figures 4.7 and 4.12. Here, we compare a candidate obtained before convergence with the converged solution obtained using NGH. The central vortex can be seen to rotate anti-clockwise throughout the orbit.

Figure 4.9a summarises the eight exact invariant solutions recovered in this study, overlaid on the probability density function (p.d.f.) of the DNS in the $(D/D_{\text{lam}}, I/D_{\text{lam}})$ plane. The darker regions indicate where the turbulent trajectory spends more time, while the coloured curves show the locations of the invariant solutions relative to this distribution. Several of the recovered solutions extend well into the high-dissipation region: in particular, the mean dissipation of R_1 , R_3 and R_4 exceeds the high-dissipation threshold $D/D_{\text{lam}} = 0.15$, with R_3 and R_4 reaching $\overline{D}/D_{\text{lam}} = 0.1905$ and 0.1928 , respectively. Thus, although high-dissipation bursts produce fewer close recurrences and are consequently more difficult to target, the recurrence-based approach nevertheless succeeded in extracting exact invariant solutions embedded within this less frequently visited region of the turbulent dynamics. These results demonstrate that high-dissipation dynamics are not inaccessible to recurrence-based invariant-solution searches, while also highlighting the need for longer trajectories and alternative search strategies to determine whether the most extreme dissipation events are associated with additional invariant solutions.

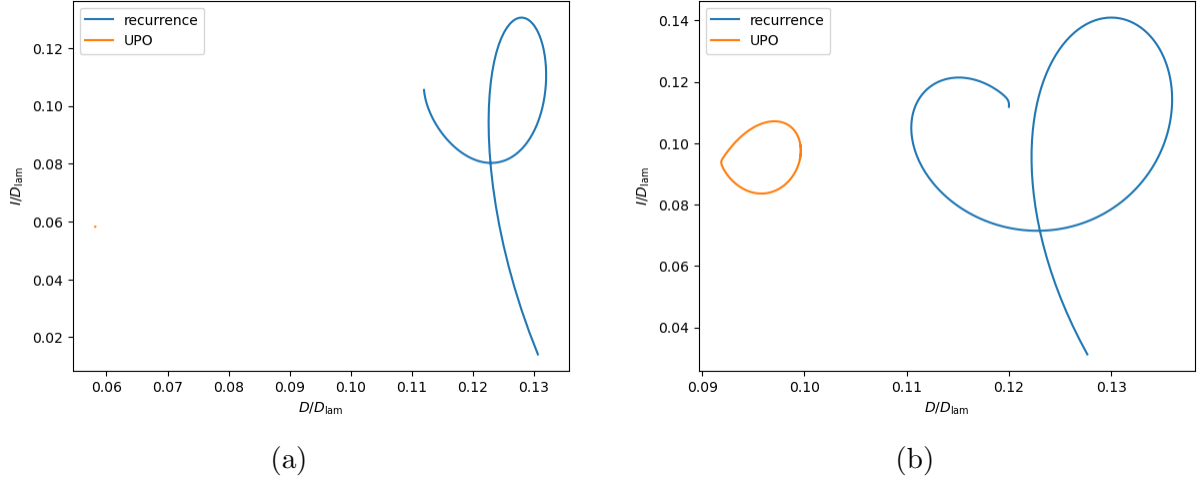


Figure 4.4: Representation of invariant solutions in the $I - D$ plane. (a) The equilibrium solution E_1 appears as a single point since the energy input and dissipation remain constant. (b) The periodic orbit P_1 traces a closed loop as the energy input and dissipation vary over one period.

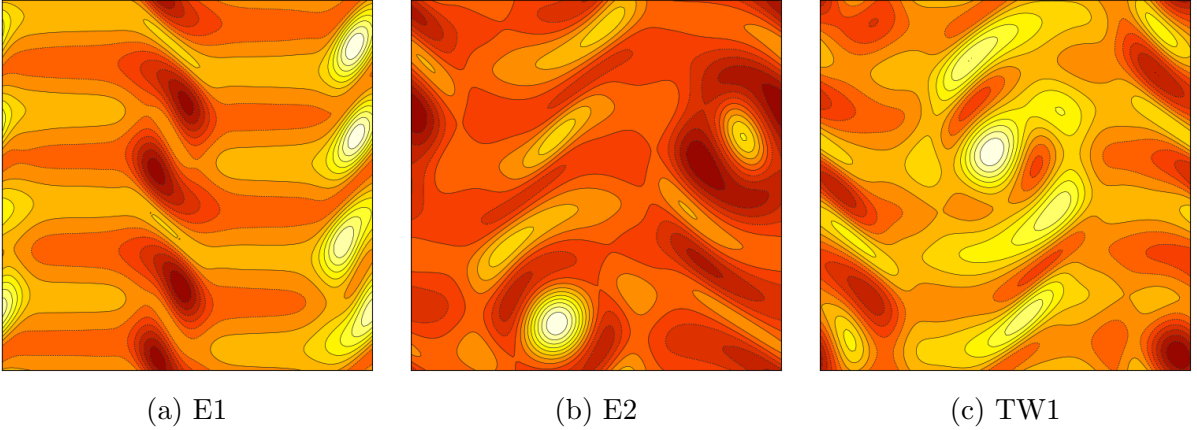
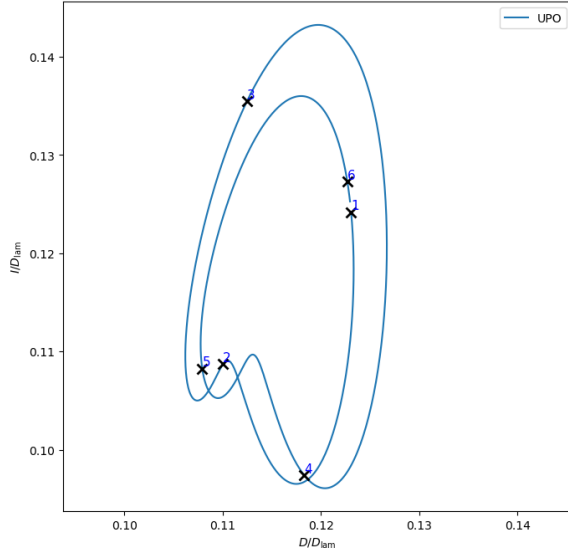
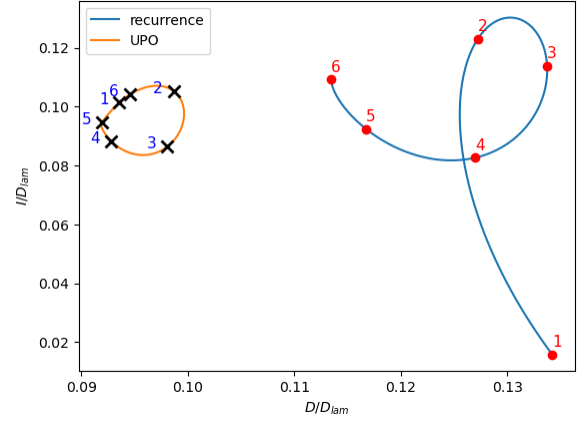


Figure 4.5: Perturbation vorticity fields for the invariant solutions at $\text{Re} = 40$, shown using 15 contour levels. The corresponding vorticity ranges are $-5.9628 \leq \omega \leq 10.9831$ for E_2 , $-5.7696 \leq \omega \leq 6.5743$ for E_1 , and $-6.5605 \leq \omega \leq 6.5605$ for TW_1 . Darker shades indicate negative vorticity, while lighter shades indicate positive vorticity.

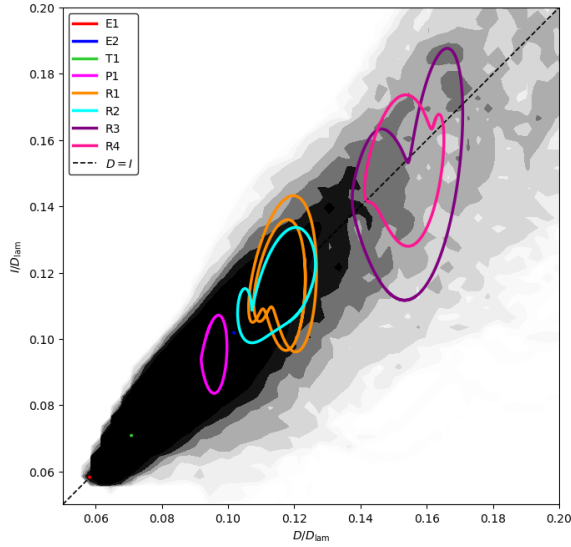


(a)

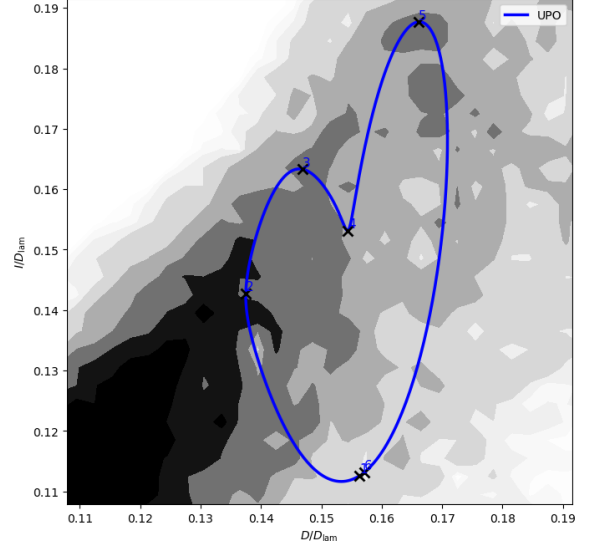


(b)

Figure 4.8: (a) $I - D$ plane for the relative periodic orbit R_1 , with the $\times$ markers indicating the states at times 0, 1.51, 3.02, 4.53, 6.04, 7.40. (b) $I - D$ plane for the periodic orbit P_1 , with the $\times$ and $\circ$ markers indicating the corresponding states at times 0, 0.56, 1.12, 1.69, 2.25, 2.82.



(a) All eight exact invariant solutions.



(b) The R_3 invariant solution

Figure 4.9: Invariant solutions in the $(D/D_{\text{lam}}, I/D_{\text{lam}})$ plane over the p.d.f. of the DNS turbulence. The p.d.f. is represented using 10 contour levels at 0.05, 0.1, $\dots$, 40, with darker shading indicating higher probability density. The dashed line denotes $D = I$.

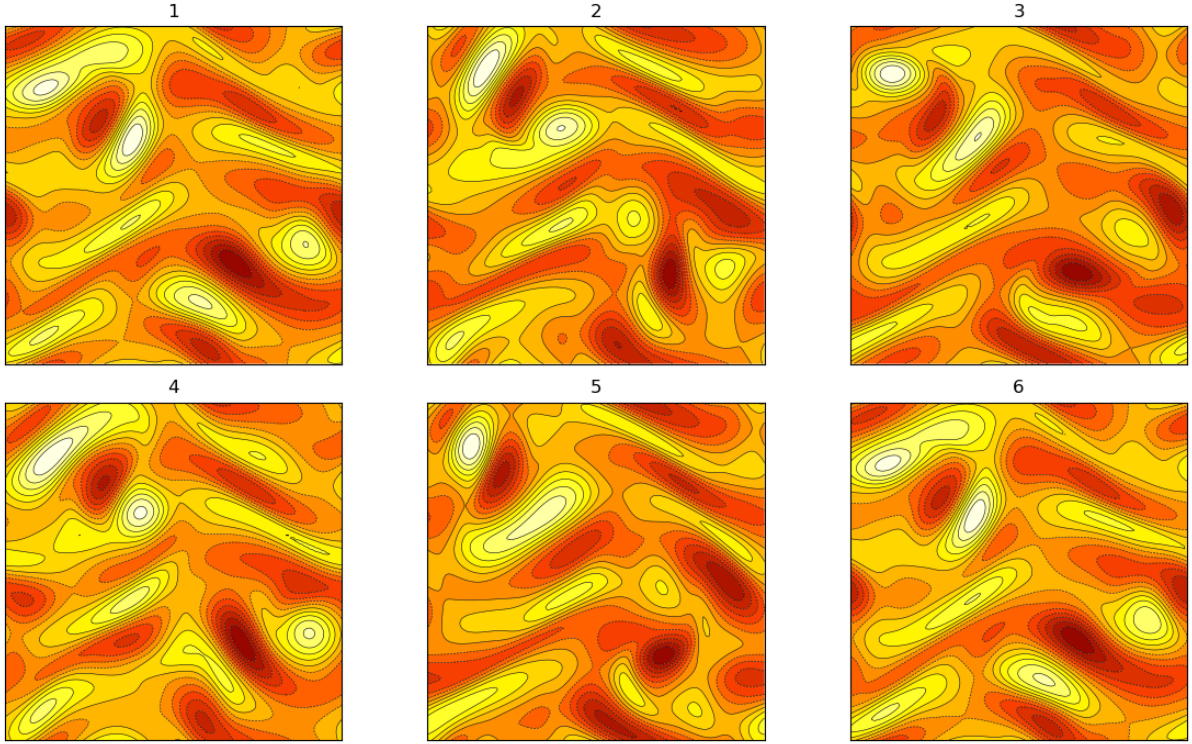


Figure 4.10: Vorticity-field snapshots of the relative periodic orbit R_1 at times 0, 1.51, 3.02, 4.53, 6.04, 7.40, shown using 15 contour levels, with $-7.68 \leq \omega \leq 7.52$. Darker shades indicate negative vorticity, while lighter shades indicate positive vorticity.

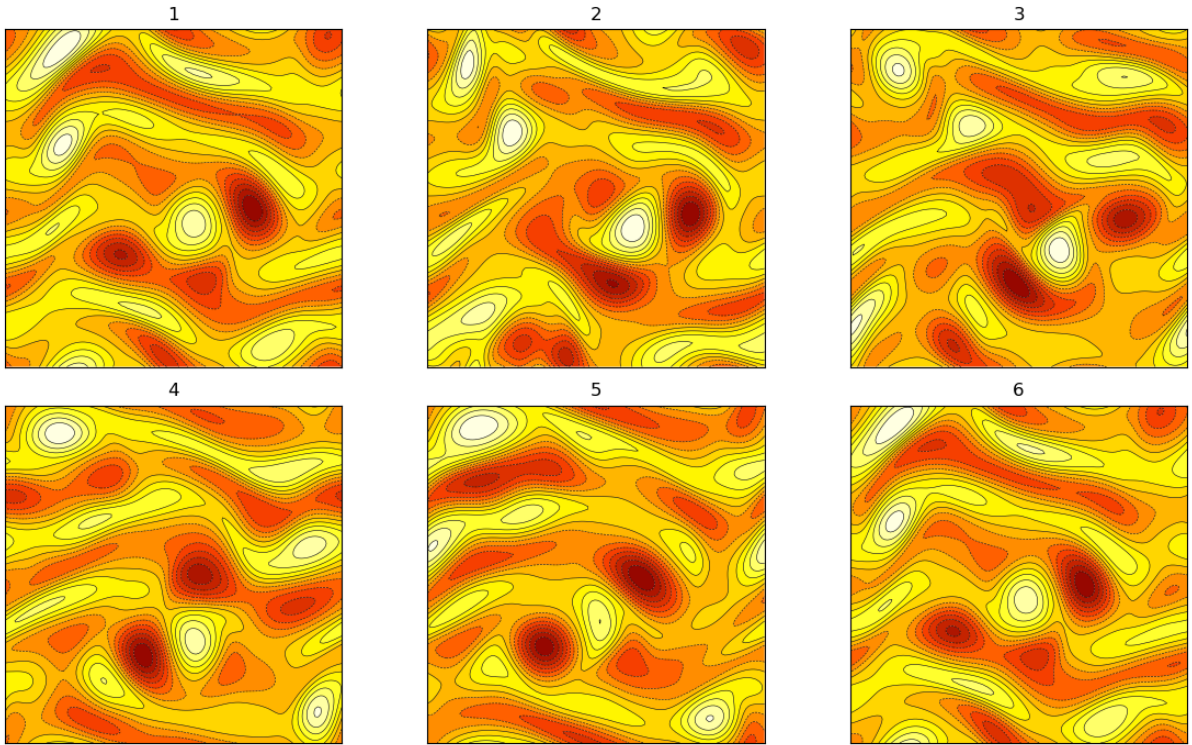


Figure 4.11: Vorticity-field snapshots of the relative periodic orbit R_3 at times 0, 1.19, 2.39, 3.58, 4.60, 5.60, shown using 15 contour levels, with $-9.56 \leq \omega \leq 7.44$. Darker shades indicate negative vorticity, while lighter shades indicate positive vorticity.

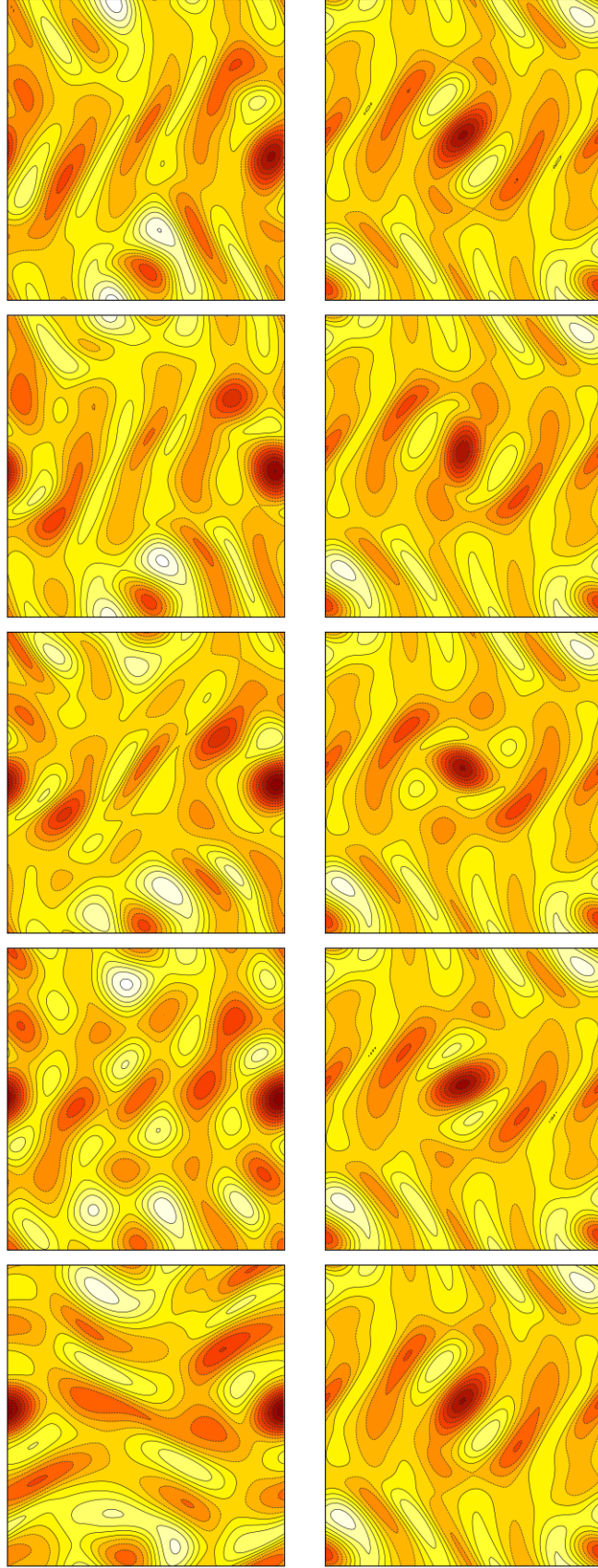


Figure 4.12: Evolution of the periodic orbit P_1 at times 0, 0.56, 1.12, 1.69, 2.25, 2.82. The left column shows the candidate solution before NGh convergence, while the right column shows the converged solution. Each panel uses 15 contour levels, with the darkest colour representing the minimum value.

Chapter 5

Summary

This work studied two-dimensional Kolmogorov flow at $\text{Re} = 40$ with forcing wavenumber $n = 4$, with particular focus on recurrent structures associated with high-dissipation episodes. A modified recurrence function, R_K , was used to identify candidate near recurrences, which were then supplied as initial guesses to the Newton–GMRES–Hookstep (NGh) method. This approach was motivated by the difficulty of identifying recurrent structures in the less frequently visited high-dissipation regions of the turbulent dynamics.

A DNS trajectory of duration 10^4 was used for the recurrence searches, with nine searches performed to investigate the effects of the recurrence and dissipation thresholds. For $D_{\text{thres}} = 0.15$ and $R_{\text{thres}} = 0.70$, the searches produced a consistent number of candidate guesses. Increasing the recurrence threshold to $R_{\text{thres}} = 0.85$ substantially increased the number of candidate guesses, but did not increase the number of converged solutions. Conversely, reducing the threshold to $R_{\text{thres}} = 0.60$ produced fewer guesses and one convergence, while increasing the dissipation threshold to $D_{\text{thres}} = 0.20$ was particularly restrictive. The combination $D_{\text{thres}} = 0.15$ and $R_{\text{thres}} = 0.70$ therefore provided the most consistent results and was used for the majority of the searches. Across all nine searches, 445 initial guesses were generated, of which 14 converged to exact invariant solutions.

An important observation was that the recurrence function R_K becomes noticeably larger during high-dissipation bursts than during the surrounding turbulent dynamics. Consequently, these regions contain fewer close recurrences from which suitable NGh initial guesses can be constructed. Despite this difficulty, the targeted recurrence search recovered invariant solutions from the high-dissipation regions, demonstrating that exact recurrent structures can still be extracted from these less frequently visited parts of the turbulent trajectory.

The 14 converged guesses corresponded to eight unique invariant solutions: two equilibria, E_1 and E_2 , a travelling wave TW_1 , a periodic orbit P_1 , and four relative periodic

orbits, R_1 – R_4 . Several of these solutions show close agreement with invariant solutions previously reported in the literature. In particular, TW_1 closely agrees with the travelling wave $\tilde{T}_1$ of Chandler and Kerswell (2013), P_1 closely matches their periodic orbit P_2 , and E_2 agrees closely with a solution reported by Farazmand (2016). These comparisons provide a useful validation of the recurrence and NGh procedure implemented in this work.

The computational analysis also revealed differences in the convergence behaviour of the NGh iterations compared with previous studies. The initial recurrence searches required approximately 1.5 hours per run, while NGh convergence represented the main computational bottleneck, with some individual guesses requiring up to one hour. Overall, the calculations were performed on the scale of hours for the recurrence searches and days for the NGh calculations, compared with the weeks and months reported by Chandler and Kerswell (2013) for recurrence detection and subsequent NGh calculations, respectively. The Newton residual histories also showed that some of the converged guesses in this work exhibited stagnation around 10^{-5} before eventually converging. This differs from the behaviour reported by Farazmand (2016), where residual stagnation above approximately 10^{-5} indicated non-convergence. Consequently, applying such a stagnation criterion as an early stopping condition would have discarded some of the successful guesses identified in this work.

Future work should first consider extending the DNS duration from 10^4 to the 10^5 time units as used by Chandler and Kerswell (2013), providing a larger sample of the turbulent dynamics and allowing it to be determined whether the poorer performance observed for $R_{\text{thres}} = 0.60$ compared with $R_{\text{thres}} = 0.70$ is a genuine effect of the threshold or a consequence of sampling variability. The number of Newton iterations could also be increased from 50 to 100 or more, since Figure 4.3 shows that our converged guesses typically required around 25–30 iterations before rapidly reducing their residual, whereas Farazmand (2016) observed rapid convergence within only a few iterations; allowing more iterations may therefore increase the number of converged solutions. Finally, although the modified recurrence function R_K is physically motivated by its greater weighting of high-wavenumber components, its advantage has not been systematically tested here. This could be investigated using DNS calculations with a fixed random seed, allowing otherwise identical recurrence searches to be compared directly. These investigations were not performed due to the time constraints of the present study.

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Appendix A

DNS Implementation and supporting

The following listing shows the pseudospectral DNS routine used to integrate the vorticity equation in time.

Listing A.1: Pseudospectral DNS routine used to integrate the vorticity equation.

```
1 from numpy.fft import fft2, ifft2, rfft2, irfft2
2 import numpy as np
3
4
5 def dns(omega0, Lx, Ly, Nx, Ny, n, Re, tend, dt):
6     omega = omega0.copy()
7
8     nx = np.arange(Nx)
9     my = np.arange(Ny//2 + 1)
10
11     nx[Nx//2 + 1:] -= Nx
12
13     # Fourier wavenumbers
14     kx = 2*np.pi*nx/Lx
15     ky = 2*np.pi*my/Ly
16
17     # spectral diffusion operator
18     K = kx[:, None]**2 + ky[None, :]**2
19     G = K / Re
20
21     denom = K.copy()
22     denom[0,0] = 1.0
23
```

```

24 def u_and_v_stream(Omega):
25     phi_hat = Omega.copy()
26     phi_hat /= denom
27     phi_hat[0,0] = 0.0
28
29     u_hat = 1j * ky[None, :] * phi_hat
30     v_hat = -1j * kx[:, None] * phi_hat
31
32     u = irfft2(u_hat, s=(Nx, Ny))
33     v = irfft2(v_hat, s=(Nx, Ny))
34
35     return u, v
36
37     # filter
38     mask = (nx[:, None]**2 + my[None, :]**2) <= (Nx / 3)**2
39
40     # forcing input
41     forcing = np.where(
42         (nx[:, None] == 0) & (my[None, :] == n),
43         n/2,
44         0.0
45     )
46
47     def f(Omega):
48         u, v = u_and_v_stream(Omega)
49         omega = irfft2(Omega, s=(Nx, Ny))
50
51         nonlinear = -1j * (
52             kx[:, None] * rfft2(u * omega)
53             + ky[None, :] * rfft2(v * omega)
54         )
55
56         nonlinear *= mask
57
58         return nonlinear + forcing
59
60     Omega = rfft2(omega)
61
62     C1 = 1.0 / (1.0 + dt * G / 2)
63     C2 = 1 - dt * G / 2
64

```

```

65     M = int(tend / dt)
66
67     for _ in range(M):
68         F = f(Omega)
69
70         Omega_pred = C1 * (
71             C2 * Omega + dt * F
72         )
73
74         Omega = C1 * (
75             C2 * Omega
76             + (dt / 2) * (f(Omega_pred) + F)
77         )
78
79     return irfft2(Omega, s=(Nx, Ny))

```

Appendix B

Recurrence Analysis

The recurrence calculation compares pairs of snapshots separated by a time lag T , while accounting for the continuous streamwise symmetry and the discrete spanwise symmetry. The main recurrence function is given in Listing B.1, with the computation over a range of time lags given in Listing B.2. The symmetry shifts are determined by Listing B.3.

Listing B.1: Recurrence function including the continuous streamwise phase shift.

```
1 def R_D_t_T_slices(t, T):
2     Omega_now = Omega_history[t]
3     Omega_old = Omega_history[t-T]
4     best = np.inf
5
6     phi_t = np.angle(Omega_now[1, 0])
7     phi_t_T = np.angle(Omega_old[1, 0])
8
9     phase_t = np.exp(-1j * KX * phi_t)
10    phase_t_T = np.exp(-1j * KX * phi_t_T)
11
12    numerator = np.sum(
13        K * np.abs(
14            Omega_now * phase_t
15            - Omega_old * phase_t_T
16        )**2
17    )
18
19    denominator = np.sum(
20        K * np.abs(Omega_now)**2
21    )
22
23    R = numerator / denominator
24    best = min(best, R)
```



```

25
26     return best

```

Listing B.2: Evaluation of the recurrence function over the range of time lags.

```

1 def compute_R(Omega_history, Tmax, dt, R_function):
2
3     Nt = len(Omega_history)
4
5     # Maximum lag in snapshots
6     max_lag = min(int(Tmax / dt), Nt - 1)
7
8     R = np.full(
9         (Nt, max_lag + 1),
10        np.nan
11    )
12
13    for i in range(Nt):
14        print(f"Processing t = {i}/{Nt-1}")
15
16        lag_max = min(max_lag, i)
17
18        for lag in range(lag_max + 1):
19            R[i, lag] = R_function(i, lag)
20
21    return R

```

Listing B.3: Determination of the continuous streamwise and discrete spanwise symmetry shifts.

```

1 def get_s_m(Omega0, Omega_T):
2
3     SxIn_Dan = np.angle(
4         Omega_T[1, 0] / Omega0[1, 0]
5     )
6
7     R_min = np.inf
8
9     for m in range(n):
10
11         phase = np.exp(
12             1j * 2*np.pi*m*KY/n
13         )

```

```

14
15     R = R_D_t_T_slices(
16         Omega0,
17         Omega_T * phase
18     )
19
20     if R < R_min:
21         R_min = R
22         m_best = m
23
24     return SxIn_Dan, m_best

```